Real Estate Quiz/Tutor Bot

Course Name: Gen AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: GAI-47

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Academic Year: 2025-26

1. Problem Statement & Objectives

1.1 Problem Statement

The real estate industry involves complex decision-making related to property valuation, zoning regulations, legal compliance, environmental considerations, and investment analysis. Students and beginners often struggle to understand practical real-world scenarios because traditional learning methods focus heavily on theory rather than applied case-based learning.

There is a need for an intelligent system that can:

- Convert real property data into learning material
- Simulate real-world analytical scenarios
- Provide interactive assessment with feedback
- Adapt difficulty based on user expertise

The challenge is to design an AI-powered educational platform that can dynamically generate contextual, property-based questions and provide structured explanations, ensuring both conceptual clarity and industry relevance.

1.2 Project Objectives

The primary objective of the Real Estate Tutor Bot is to develop an AI-powered educational system that transforms real-world property data into interactive learning experiences.

The key objectives of the project are:

- To design a system that accepts real estate property listings in structured JSON format.
- To implement a Retrieval-Augmented Generation (RAG) architecture for context-aware question generation.
- To generate dynamic multiple-choice questions (MCQs) based on uploaded property data.
- To provide intelligent answer evaluation with detailed explanations and industry insights.
- To personalize learning based on user-selected experience level (Beginner, Intermediate, Experienced).
- To ensure modular, scalable, and extensible system design.

1.3 Scope of the Project

The scope of the Real Estate Tutor Bot includes the development of an AI-based educational web application that:

- Accepts property listings in JSON format from users.
- Converts structured property data into vector embeddings.
- Stores embeddings in a vector database (ChromaDB).
- Retrieves relevant property data using similarity search.
- Generates contextual MCQs using Gemini 2.5 Flash.
- Evaluates user responses and provides structured feedback.

The project is limited to an academic prototype implementation using Streamlit and does not include user authentication, large-scale deployment, or production-level infrastructure.

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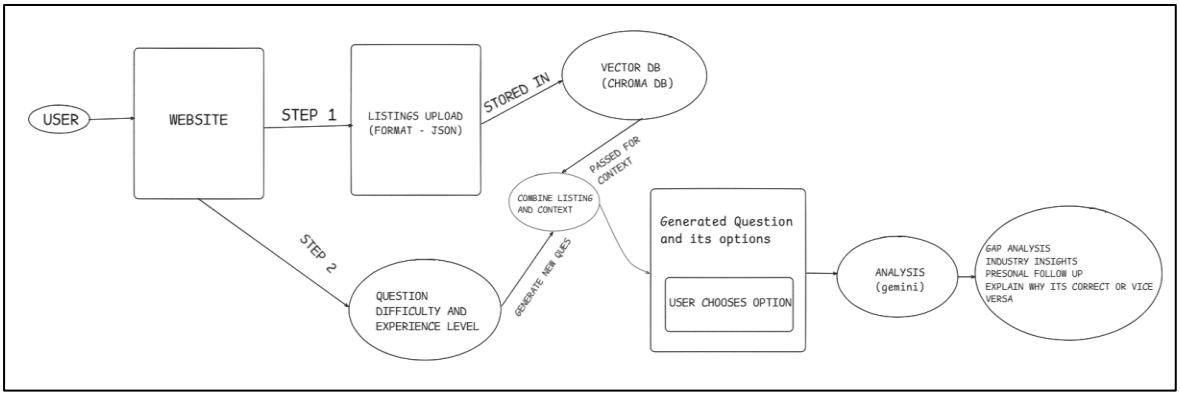
2. Proposed Solution

2.1 Key features

- Upload property listings in structured JSON format.
- Automatic validation and preprocessing of uploaded data.
- Generation of vector embeddings using the 001 embedding model.
- Storage and retrieval of embeddings using ChromaDB (Vector Database).
- Context-aware MCQ generation using Gemini 2.5 Flash.
- Adaptive learning based on experience level (Beginner, Intermediate, Experienced).
- Customizable response format (Concise or Detailed).
- Automated answer evaluation with explanation and industry insights.
- Suggestion of follow-up study topics.
- Modular and extensible system architecture.

2.2 Overall Architecture / Workflow



The Real Estate Tutor Bot follows a Retrieval-Augmented Generation (RAG) architecture.

The workflow begins when a user uploads property listings in JSON format through the Streamlit interface. The system validates and parses the data, converts relevant property information into textual format, and generates vector embeddings using the 001 embedding model. These embeddings are stored in ChromaDB along with associated metadata.

When the user selects experience level and response type and clicks "Generate Question," the system retrieves relevant property data using semantic similarity search. The retrieved context, combined with user preferences, is sent to Gemini 2.5 Flash, which generates a multiple-choice question.

After the user selects an answer, the system sends the response back to Gemini for evaluation. The AI model then provides structured feedback, including explanations, industry insights, and recommended follow-up topics.

2.3 Tools & Technologies Used

- **Frontend & Backend:** Streamlit (Python-based web framework)
- **Programming Language:** Python
- **AI Model:** Gemini 2.5 Flash
- **Embedding Model:** 001 Embedding Model
- **Vector Database:** ChromaDB
- **Architecture Pattern:** Retrieval-Augmented Generation (RAG)
- **Diagram Tool:** PlantUML

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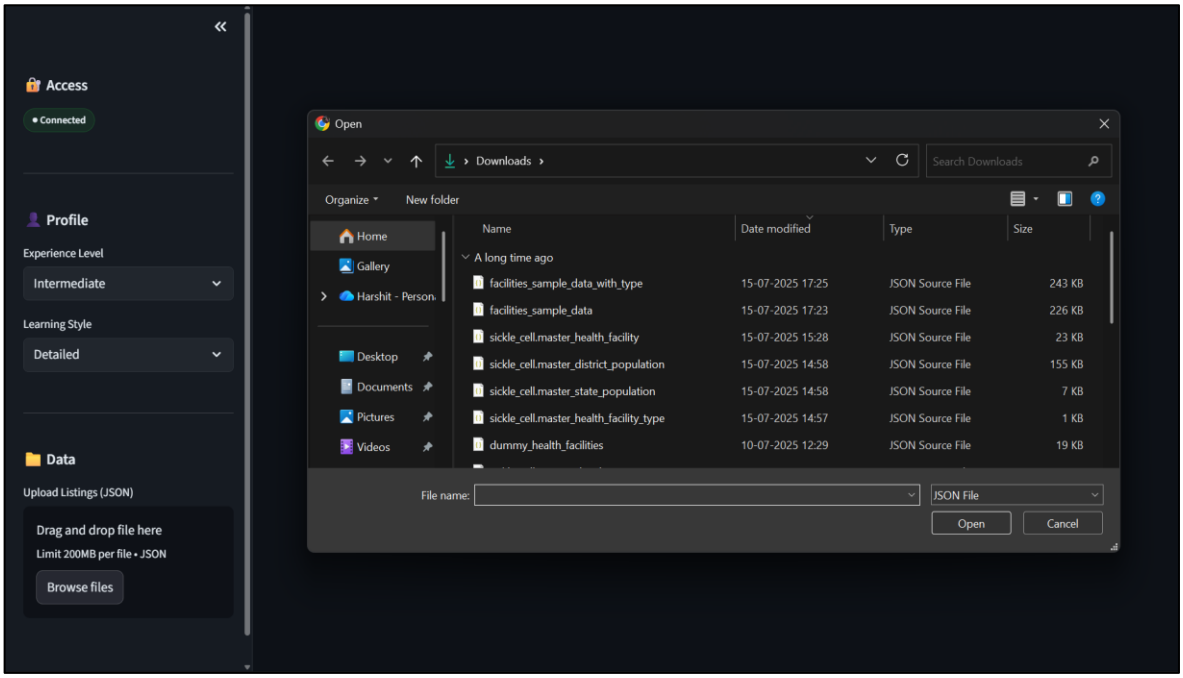
3. Results & Output

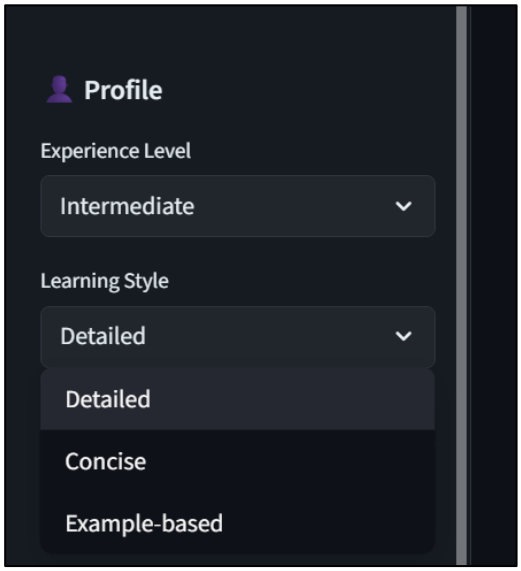
3.1 Screenshots

The developed system successfully generates AI-based multiple-choice questions from user-uploaded property listings. The key outputs of the system include:

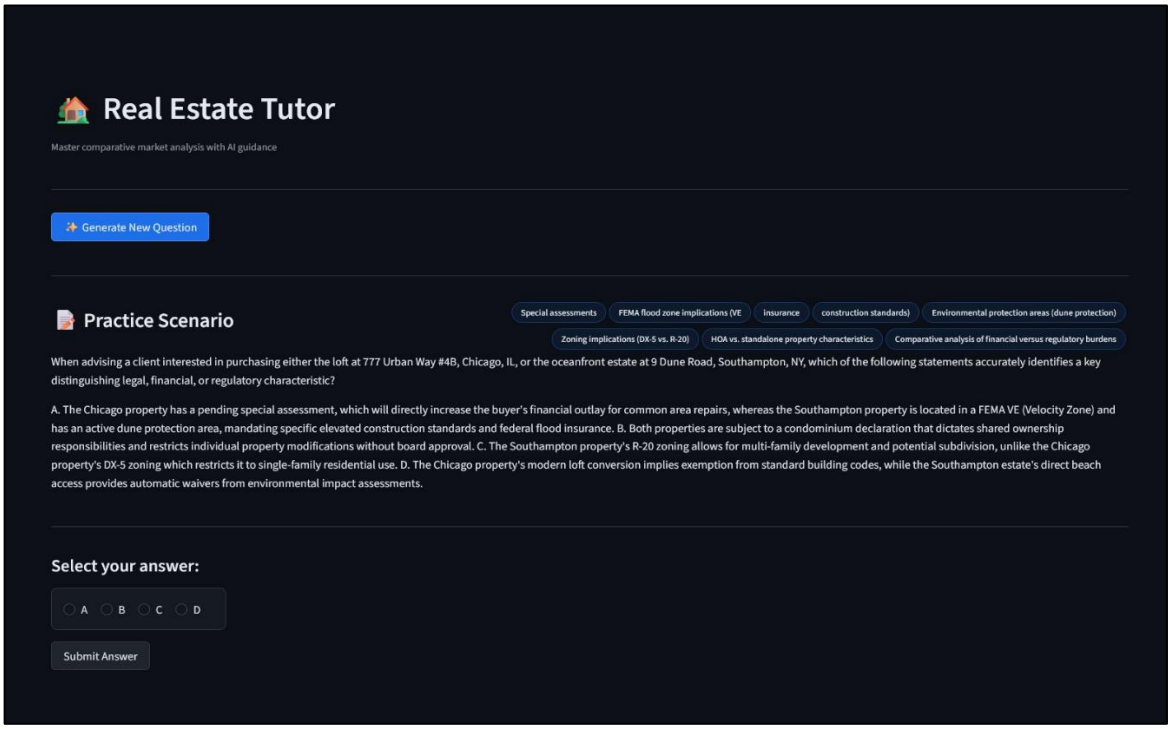
- Property JSON upload interface in Streamlit



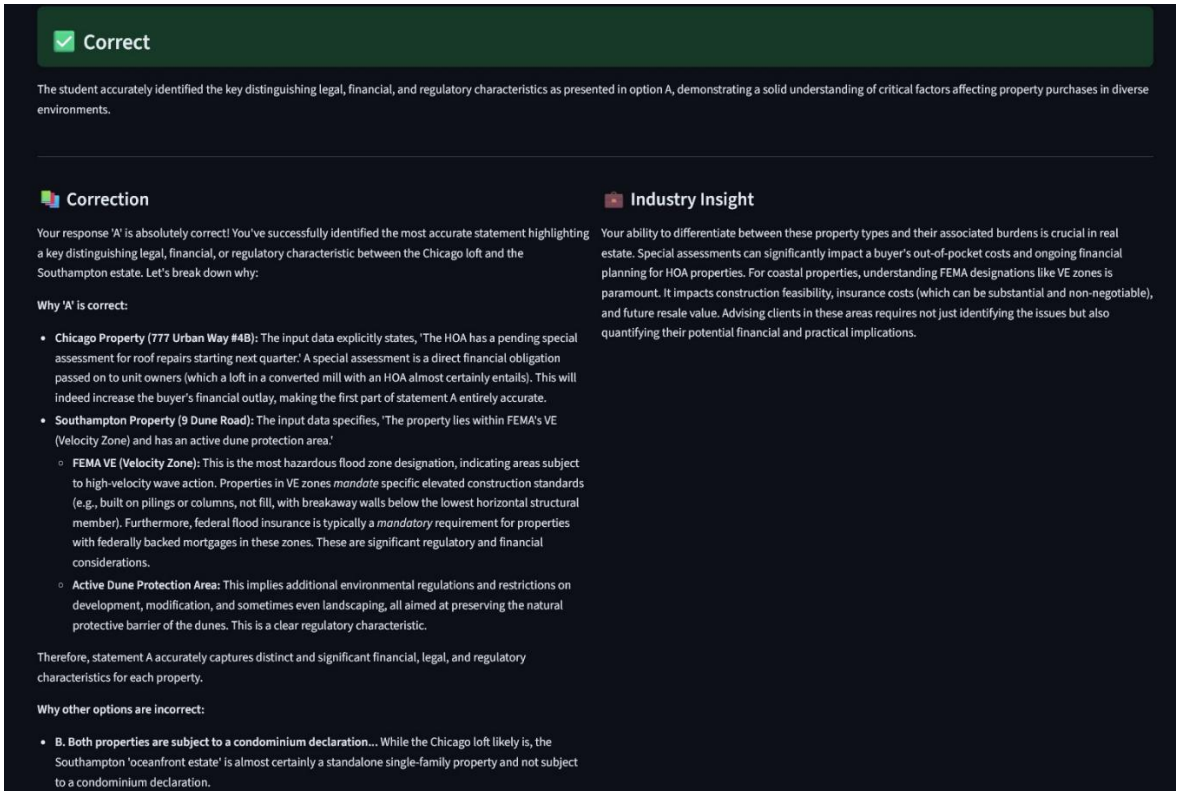
- Experience level and response type selection interface



- Dynamically generated MCQ questions



- Detailed explanation with industry insights



- Follow-up study topic recommendations

3.2 Models

The system integrates the following core models and components:

- Gemini 2.5 Flash model for question generation and evaluation
- 001 Embedding model for vector representation
- ChromaDB vector database for semantic retrieval

Although the current implementation does not include analytical dashboards, the architecture supports future integration of performance tracking and reporting features such as user progress analytics and scoring summaries.

3.3 Key Outcomes

- Successful implementation of a Retrieval-Augmented Generation (RAG) architecture.
- Accurate context-aware MCQ generation based on real property data.
- Intelligent answer evaluation with structured explanations.
- Personalized learning experience based on difficulty selection.
- Modular and scalable system design suitable for future deployment.

4. Conclusion

The Real Estate Tutor Bot successfully demonstrates the integration of Artificial Intelligence with real-world real estate education. By implementing a Retrieval-Augmented Generation (RAG) architecture, the system is capable of transforming structured property listings into dynamic, context-aware multiple-choice questions.

The project effectively combines Streamlit, vector embeddings, ChromaDB, and Gemini 2.5 Flash to create an interactive and adaptive learning platform. It provides personalized learning experiences based on user-selected difficulty levels and generates structured feedback including explanations and industry insights.

Through this project, we gained practical understanding of AI integration, vector databases, prompt engineering, and modular system design. The implementation highlights how AI can be leveraged to enhance domain-specific education using real data.

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5. Future Scope & Enhancements

The current system is an academic prototype and offers several opportunities for future enhancement:

- Implementation of user authentication and profile management.
- Tracking user performance and generating progress analytics dashboards.
- Deployment on cloud platforms (AWS / GCP / Azure) for scalability.
- Optimization of caching strategies to reduce API costs.
- Support for additional property attributes such as tax records, ownership history, and valuation metrics.
- Integration of adaptive difficulty adjustment based on user performance.

These enhancements can transform the prototype into a full-scale intelligent real estate learning platform suitable for professional training and industry applications.